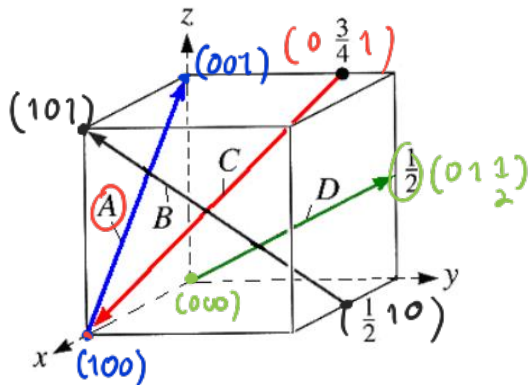


Assignment 1

316-221 Metallurgy

Arrangement of Atoms and Crystal Structure

1. Determine the Miller indices for the direction shown in the following cubic unit cell. (4 points)



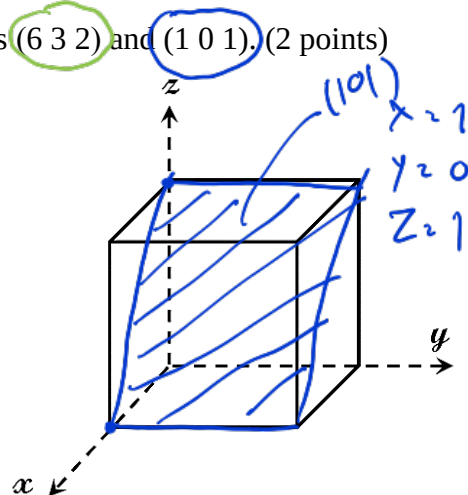
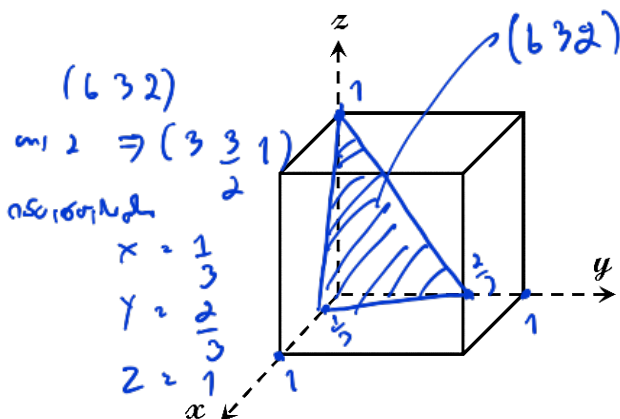
① $001 - 100 = [-101]$ ✓ 1 point

② $101 - \frac{1}{2}10 = [\frac{1}{2} - 1 1] \Rightarrow [1 - 2 2]$ ✓ 1 point

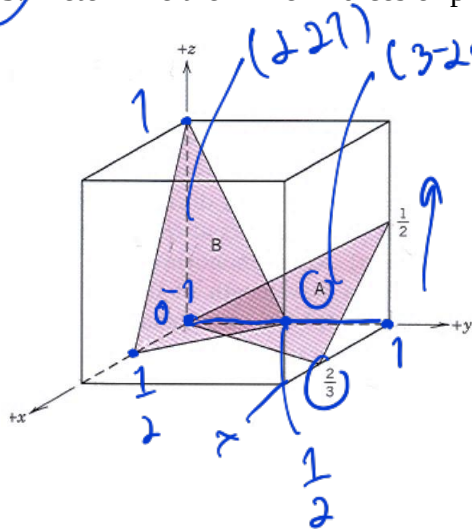
③ $100 - 0\frac{3}{4}1 = [1 - \frac{3}{4} - 1] \Rightarrow [4 - 3 - 4]$ ✓ 1 point

④ $01\frac{1}{2} - 000 = [0 1 \frac{1}{2}] \Rightarrow [0 2 1]$ ✓

2. Construct planes by Miller indices of planes (6 3 2) and (1 0 1). (2 points)



3. Determine the Miller indices of planes. (2 points)



① $x = \frac{1}{2}, y = 1, z = 1$

$x = \frac{1}{2}, y = 1, z = 1$

$2(\frac{1}{2} - 1 1) \Rightarrow (1 - 2 2) \times \checkmark$

② $x = \frac{1}{2}, y = \frac{1}{2}, z = 1$

$x = 2, y = 2, z = 1$

$(2 2 1)$